

Opgave 31

Def: $\|x\|_\infty = \max_k \{ |x_k| \}$ ← maximum largest absolute value of

component in vector

$$\|v\|_\infty = |\lambda| \|v\|$$

we index largest element in v with k (and multiply by $|\lambda| = 1$ which we assume is 1)

$$|\lambda| |v_k| = |\lambda v_k|, \text{ trivial}$$

$$|\lambda v_k| = \left| \sum_{j=1}^n a_{kj} v_j \right| \text{ is true because } Av_k = \sum_{j=1}^n a_{kj} v_j$$

↑ and here we just add abs

$$\left| \sum_{j=1}^n a_{kj} v_j \right| \leq \sum_{j=1}^n |a_{kj} v_j| \leftarrow \text{triangle inequality}$$

$$\sum_{j=1}^n |a_{kj} v_j| = \sum_{j=1}^n a_{kj} |v_j|$$

↑
always positive, as per definition of A

$$\sum_{j=1}^n a_{kj} |v_j| \leq \sum_{j=1}^n a_{kj} \|v\|_\infty$$

part 2

cause

$$|v_j| \leq \|v\|_\infty \text{ by definition}$$

Opgave 31 part 2

$$\sum_{j=1}^n a_{kj} \|v\| = \|v\|$$

must be true cause every row in A sums to 1, thus $1 \cdot \|v\| = \|v\|$

So we get that

$\|v\| \leq \|v\|$, thus the expression is bound from both sides by the same value

We also have that $|v_j| = \|v\|_\infty$ for $\forall j$
and that $\text{Arg}(v_j) = \text{Arg}(v_i)$ for $\forall j, i$

hence: $v = c(1, 1 \dots 1)$

to remove $\lambda = -1$:

$$\lambda v = \lambda v, \quad A e = e$$

$$\Downarrow \quad A e = \lambda e, \quad \underline{\underline{\lambda = 1}} \quad \left(\begin{array}{l} -1 \text{ is not possible} \\ \text{cause then we get} \\ A v = -v, \text{ not true} \end{array} \right)$$